**LAB MANUAL**
**LIST OF EXPERIMENTS**

1. Study of Logic gates.
2. Verify four voltage and current parameters for TTL and CMOS (IC 74LSXX, 74HCXX), (Refer Data-Sheet).
3. Study of IC-74LS153 as a Multiplexer. (Refer Data-Sheet).
   - Design and Implement 8:1 MUX using IC-74LS153 & Verify its Truth Table.
   - Design & Implement the given 4 variable function using IC74LS153. Verify its Truth-Table.
4. Study of IC-74LS138 as a Demultiplexer/ Decoder (Refer Data-Sheet).
   - Design and Implement full adder and subtractor function using IC-74LS138.
   - Design & Implement 3-bit code converter using IC-74LS138.(Gray to Binary/Binary to Gray)
5. Study of IC-74LS83 as a BCD adder,(Refer Data-Sheet).
   - Design and Implement 1 digit BCD adder using IC-74LS83
   - Design and Implement 4-bit Binary adder and subtractor using IC-74LS83.
6. Study of IC-74LS85 as a magnitude comparator,(Refer Data-Sheet)
   - Design and Implement 4-bit Comparator.
   - Design and Implement 8-bit Comparator
7. Study of Counter ICs (74LS90/74LS93). (Refer Data-Sheet)
   - Design and Implement MOD-N and MOD-NN using IC-74LS90 and draw Timing Diagram.
   - Design and Implement MOD-N and MOD-NN using IC-74LS93 and draw Timing Diagram.
8. Study of synchronous counter
   - Design & Implement 4-bit Up/down Counter and MOD-N Up/down Counter using IC-74HC191/IC74HC193. Draw Timing Diagram
9. Study of Shift Register (74HC194/74LS95), (Refer data-Sheet)
   - Design and Implement Pulse train generator using IC-74HC194/IC74LS95 (Use right shift/left shift).
   - Design and Implement 4-bit Ring Counter/ Twisted ring Counter using shift registers IC 74HC194/IC74LS95.
10. Write, simulate and verify, VHDL Code for four bit logical and arithmetic operations for ALU.
    - Behavioral modeling
    - Dataflow modeling
11. D FF and JK FF (With Synchronous and asynchronous reset input) (Use Behavioral modeling)
    - Write, simulate and verify, VHDL Code for D flip flop using Synchronous /asynchronous reset input
    - Write, simulate and verify, VHDL Code for JK flip flop using asynchronous set /reset Input

12. Four bit ripple counter**.** (Use data flow/Structural modeling)

- Write, simulate and verify, VHDL code for four bit ripple up counter
- Write, simulate and verify VHDL code for four bit ripple up/down Counter using mode control.

## EXPERIMENT NO.1

## STUDY OF LOGIC GATES

**AIM :** To study about logic gates and verify their truth tables.

**APPARATUS REQUIRED:**

| SL No. | COMPONENT | SPECIFICATION | QTY |
|---|---|---|---|
| 1. | AND GATE | IC 7408 | 1 |
| 2. | OR GATE | IC 7432 | 1 |
| 3. | NOT GATE | IC 7404 | 1 |
| 4. | NAND GATE 2 I/P | IC 7400 | 1 |
| 5. | NOR GATE | IC 7402 | 1 |
| 6. | X-OR GATE | IC 7486 | 1 |
| 7. | X-NOR GATE | IC 74266 | 1 |
| 8. | IC TRAINER KIT | - | 1 |
| 9. | PATCH CORD | - | 30 |

**THEORY:**

Circuit that takes the logical decision and the process are called logic gates. Each gate has one or more input and only one output.

OR, AND and NOT are basic gates. NAND, NOR and X-OR are known as universal gates. Basic gates form these gates.

**AND GATE:**

The AND gate performs a logical multiplication commonly known as AND function. The output is high when both the inputs are high. The output is low level when any one of the inputs is low.

**OR GATE:**

The OR gate performs a logical addition commonly known as OR function. The output is high when any one of the inputs is high. The output is low level when both the inputs are low.

**NOT GATE:**

The NOT gate is called an inverter. The output is high when the input is low. The output is low when the input is high.

**NAND GATE:**

The NAND gate is a contraction of AND-NOT. The output is high when both inputs are low and any one of the input is low .The output is low level when both inputs are high.

**NOR GATE:**

The NOR gate is a contraction of OR-NOT. The output is high when both inputs are low. The output is low when one or both inputs are high.

**X-OR GATE:**

The output is high when any one of the inputs is high. The output is low when both the inputs are low and both the inputs are high.

**PROCEDURE:**

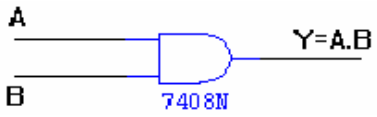
(i) Connections are given as per circuit diagram.

(ii) Logical inputs are given as per circuit diagram.

(iii) Observe the output and verify the truth table.

**AND GATE:**

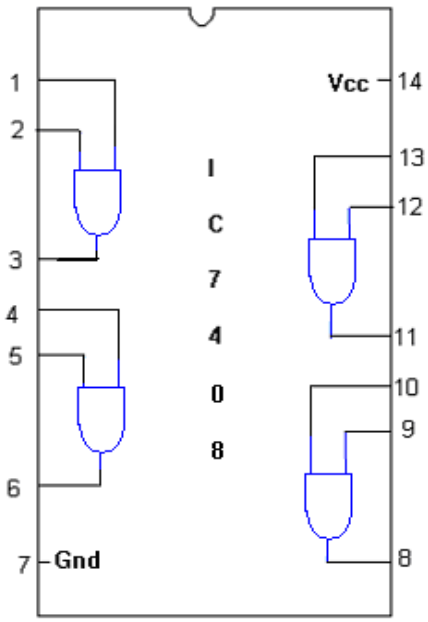
**SYMBOL:**




TRUTH TABLE

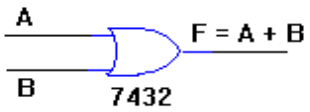
| A | B | A.B |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 0 |
| 1 | 0 | 0 |
| 1 | 1 | 1 |

**PIN DIAGRAM:**

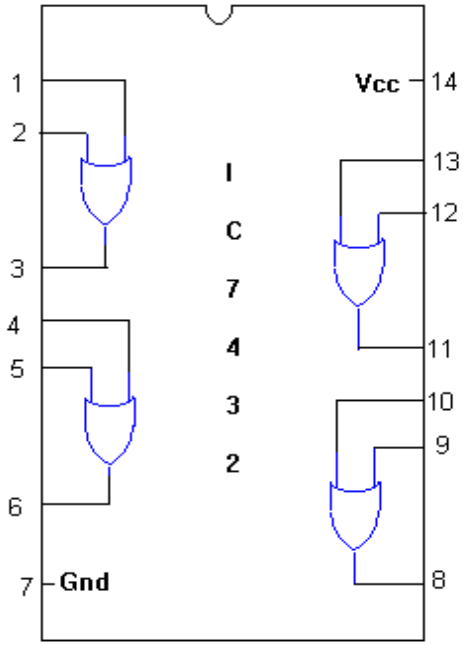



**OR GATE:**

**SYMBOL:**




**PIN DIAGRAM:**

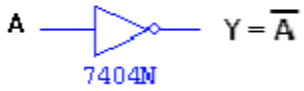



TRUTH TABLE

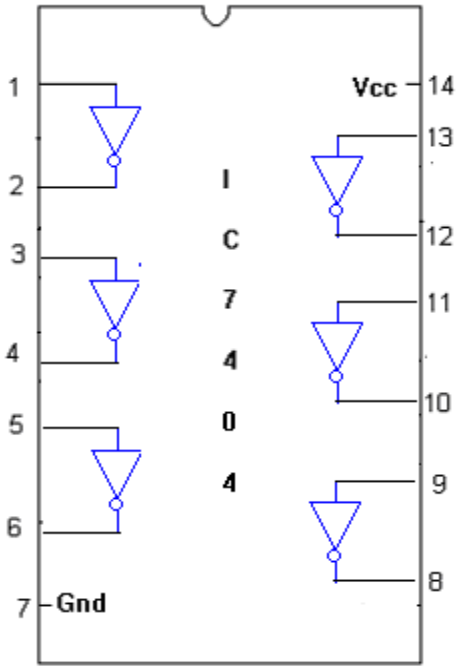
| A | B | A+B |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 1 |

**NOT GATE:**

**SYMBOL:**




**PIN DIAGRAM:**

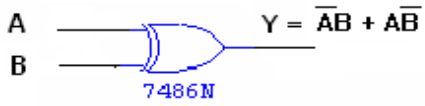



TRUTH TABLE :

| A | $\overline{A}$ |
|---|---|
| 0 | 1 |
| 1 | 0 |

**X-OR GATE :**

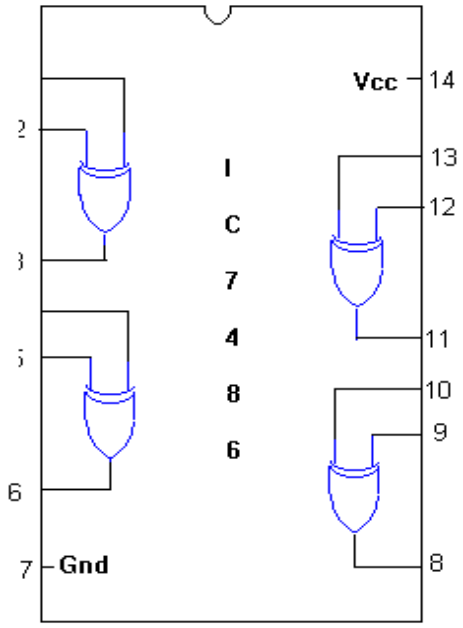
**SYMBOL :**




TRUTH TABLE :

| A | B | $\overline{A}B + A\overline{B}$ |
|---|---|---|
| 0 | 0 | 0 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

**PIN DIAGRAM :**




**2-INPUT NAND GATE:**

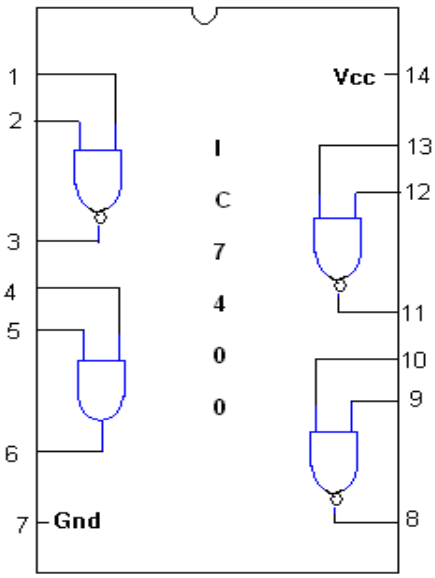
**SYMBOL:**




TRUTH TABLE

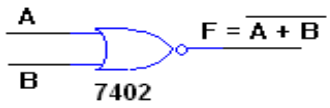
| A | B | $\overline{A \cdot B}$ |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

**PIN DIAGRAM:**

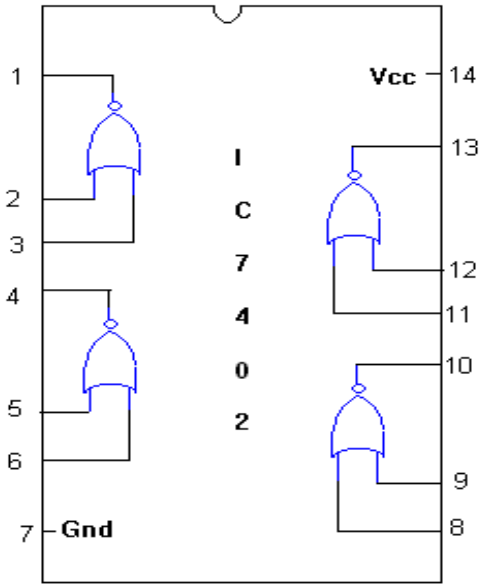



**NOR GATE:**

**SYMBOL :**




**PIN DIAGRAM :**




**TRUTH TABLE**

| A | B | $\overline{A+B}$ |
|---|---|---|
| 0 | 0 | 1 |
| 0 | 1 | 1 |
| 1 | 0 | 1 |
| 1 | 1 | 0 |

**CONCLUSION:**

## EXPERIMENT NO.2

## STUDY OF TTL AND CMOS IC FAMILIES.

**AIM :** To verify four voltage and current parameters for TTL and CMOS .

**THEORY:**

A group of compatible IC's with the same logic levels and supply voltages for performing various logic functions have been fabricated using a specific circuit configuration which is referred to as logic family.

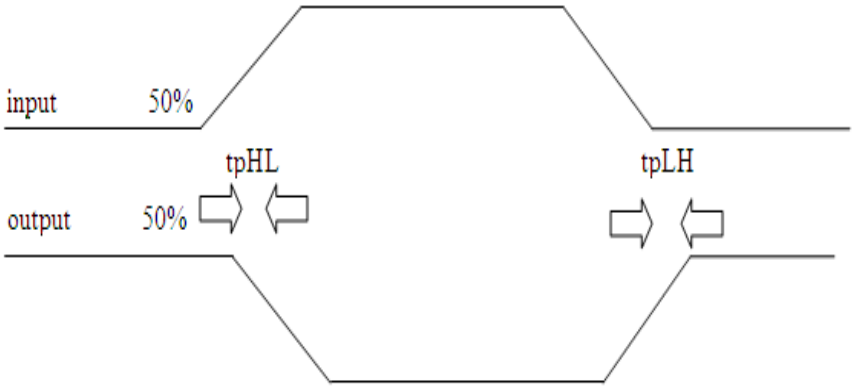
### I) Characteristics of TTL IC's:

The various characteristics of digital ICs used to compare their performances are

1. Speed of operation
2. Power dissipation
3. Figure of merit
4. Fan out
5. Current and voltage
6. Noise immunity
7. Operating temperature range
8. Power supply requirements
9. Flexibilities available

### 1. Speed of operation:

The speed of digital circuits is specified in terms of the propagation delay time. The propagation delay time is the time interval between the application of input voltage and occurrences of resultant o/p voltage and is measured between 50% voltage levels of input and output waveforms . There are two delay times: t pHL =when output is changing from HIGH state to LOW state. tpLH= when output is changing from LOW state to HIGH state. The propagation delay time is taken as the average of these two delay times.




The propagation delay time is expressed in terms of :

1)tpHL (time it takes for change in the input to cause a HIGH to LOW transition in the output)

2)tpLH(time it takes for change in the input to cause a LOW to HIGH transition in the output)

Standard TTL has maximum tpHL of 15ns and maximum tpLH of 22ns.

**2. Power dissipation**

This is the amount of power dissipated in an IC. It is given by the product of Vcc and Icc,where Icc is the amount of current drawn from the supply Vcc. Icc is the average value of Icc(0) & Icc(1). This power is specified in milliwatts.

i.e. Icc(avg)=(Icc(0)+Icc(1))/2

where Icc(0)= the value of Icc drawn from Vcc when output is low.

Icc(1)=the value of Icc drawn from Vcc when output is high.

Pd=Vcc*Icc(avg)

**3. Figure of merit:**

The fig. of merit of digital IC is defined as the product of speed and power. The speed is specified in terms of propagation delay time expressed in nanoseconds.

Figure of merit=propagation delay time(ns)*power(mW)

It is specified in pico joules(pJ) (ns*mW=pJ).A low value of speed power product is desirable.

**4. Fan out**

This is the no. of similar gates which can be driven by a gate . Fan out is the measure of no. of loads that a gate can drive .High fan out is advantageous because it reduces the need of additional drivers to drive more gates.

$$\text{Fan out} = \frac{\text{I(ol) of the driving gate}}{\text{I(il) of the driven gate}}$$

Where I(ol) and I(il) are low level output and low level input current respectively, as explained below.

5. **Current and voltage parameters**:

High level input voltage (Vih): this is the minimum input voltage which is recognized by the gate as logic 1.

Low level input voltage (Vil): this is the maximum input voltage which is recognized by the gate as logic 0.

High level output voltage (Voh): this is the minimum voltage available at the output corresponding to logic 1.

Low level output voltage(Vol): this is the maximum voltage available at the output corresponding to logic 0.

High level input current (Iih): this is the minimum current which must be supplied by a driving source corresponding to logic 1 level voltage.

Low level input current (Ioh): this is the minimum current which must be supplied by driving source corresponding to logic 0 level voltage.

High level output current (Ioh): this is the maximum current which the gate can sink in 1 level.

Low level output current (Iol): this is the maximum current which the gate can sink in 0 level.

High level supply current , Icc (1): this is the supply current when the output of the gate is at logic 1.

Low level supply current , Icc(0): this is the supply current when the output of the gate is at logic 0.

Calculations for fan out:

- Sinking current is the conventional current flowing into the gate and the gate is said to be sinking current .

- Sourcing current is the conventional current flowing out of the gate and the gate is said to be ‘sourcing ’ or supplying current.

A TTL device can source current (high output) or sink current (low output).Data sheets of standard TTL devices indicate that any 7400 series device can sink upto 16mA, designated ,

I(ol)max=16mA

And can source upto 400uA,designated,

I(oh)max=-400uA(minus sign means that the conventional current is out of device, and a plus sign means it is into the device.)

The worst case TTL input currents are:

I(il)max=-1.6mA I(ih)=4

A 7400 NAND gate sinking current from ten other gates, each with low level input current of -1.6mA. The 1.6mA is said to be ‘one standard TTL load’. Fan out is a measure of the number of loads that a gate can drive.

$$\text{Fan out} = \frac{I(ol) \text{ of the driving gate.}}{I(il) \text{ of the driven gate}}$$

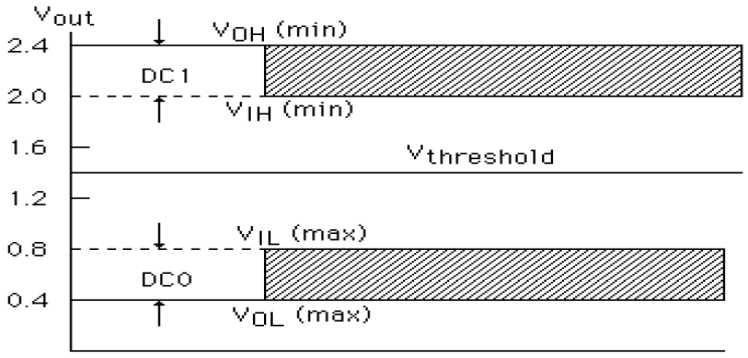
For a NAND gate driving other NAND gates or inverters,

$$\text{Fan-out} = \frac{I(ol)}{I(il)} = \frac{16mA}{1.6mA} = 10 \text{ standard loads}$$

Therefore, each of the standard TTL gate can drive 10 other standard gates.

6. **Noise immunity**:

Stray electric and magnetic fields may induce unwanted voltages , known as noise , on the connecting wires between logic circuits. This may cause the voltage at the input to a logic circuit to drop below Vth or rise above V(IL) and may produce undesired operation .The circuit’s ability to tolerate noise signals is referred to as the noise immunity, a quantitative measure of which is called as noise margin.




Input and Output TTL Voltage Levels Illustrating DC Noise Margin.

**7. Operating temperature**

The temperature range in which an IC functions properly is specified. The accepted temp ranges are: 0 to +70C for consumer and industrial applications and -55 to +125 C for military purposes.

**8. Power supply requirements**

The supply voltages and amount of power required by an IC are important characteristics required to choose the proper power supply.

**9. Flexibilities available**

Various flexibilities are available in different IC logic families and these must be considered while selecting a logic family for a particular job. Some of the flexibilities available are:

1. The breadth of the series: types of different logic functions available in series.
2. Popularity of the series: the cost of manufacturing depends upon no. of IC's manufactured. When a large number of IC's of one type are manufactured, the cost per function will be very small and it will be easily available because of multiple sources.
3. Wired logic capability: the outputs can be connected together to perform additional logic without any extra hardware.
4. Availability of complement outputs: this eliminates the need of additional inverters.
5. Type of output: passive pull-up, active pull-up, open collector /drain, and tristate.

**II) Transistor transistor Logic (TTL)**

IC's are identified by a number code stamped on the top . The prefix is a manufacturer's code. The next two numbers denote the family of IC's such as TTL or CMOS .If letters follow, they indicate the subfamily of the IC. The next numbers indicate the function of the IC, and the last letters indicate the package style.

TTL refers to the technology for designing and fabricating digital integrated circuits that employ logic gates consisting primarily of bipolar transistors. TTL is named for its dependence on transistor alone to perform basic logical operations.

TTL IC's with numbering scheme:

| Sr. no. | series | prefix | examples |
|---|---|---|---|
| 1 | Standard TTL | 74- | 7402,74193 |
| 2 | High power TTL | 74H- | 74H02,74H193 |
| 3 | Low power TTL | 74L- | 74L02,74L193 |
| 4 | Schottky TTL | 74S- | 74S02,74S193 |
| 5 | Low power schottky TTL | 74LS- | 74LS02,74LS193 |
| 6 | Advanced Schottky TTL | 74AS- | 74AS02,74AS193 |
| 7 | Advanced Low power schottky TTL | 74ALS- | 74ALS02,<br>74 ALS 193 |

| Sr. no. | series | prefix | examples |
|---|---|---|---|
| 1 | CMOS | 74C | 74C02,74C193 |
| 2 | High Speed CMOS | 74HC | 74H02,74H193 |
| 3 | Advanced CMOS | 74AC | 74L02,74L193 |
| 4 | Advanced High Speed CMOS | 74AHC | 74S02,74S193 |
| 5 | Low Voltage CMOS | 74LVC | 74LS02,74LS193 |

**VI) comparison of TTL and CMOS logic families:**

| Sr. no. | parameter | TTL | CMOS |
|---|---|---|---|
| 1 | Operating temperature range | Military:54,54L,54LS<br>-55 to 125 C | |

| | | | |
|---|---|---|---|
| | | General: 74,74L,74LS<br>0 to 70 C | |
| 2 | technologies | Bipolar junction (here, transistor are current operated, so lower no of gates can be put on an IC ) | Field effect(here, transistors are voltage operated devices, so result in a much higher density of logic gates that can be put on an Ic.) |
| 3 | Power dissipation | High power dissipation (10mW/gate) | Low power dissipation(0.01mW/gate) |
| 4 | Fan out | 10 | 50 |
| 5 | Propagation delay | 10nS/gate | 70nS /gate |
| 6 | Speed of operation | Fastest logic family | Comparatively slow |
| 7 | Static electricity | Much less sensitive to damage from electrostatic discharge | More susceptible to damage from electrostatic discharge |

**CONCLUSION:**

## EXPERIMENT NO.3

## DESIGN AND IMPLEMENTATION OF MULTIPLEXER USING 74153

**AIM:** Study of IC-74LS153 as a Multiplexer.

1. Design and Implement 8:1 MUX using IC-74LS153 & Verify its Truth Table.
2. Design & Implement the given 4 variable function using IC74LS153. Verify its Truth-Table1.

**APPARATUS REQUIRED:**

| Sl.No. | COMPONENT | SPECIFICATION | QTY. |
|---|---|---|---|
| 1. | MULTIPLEXER | IC 74153 | 2 |
| 2. | NOT GATE | IC 7404 | 1 |
| 3. | OR GATE | IC 7432 | 1 |
| 3. | IC TRAINER KIT | - | 1 |
| 4. | PATCH CORDS | - | 32 |

**THEORY:**

**MULTIPLEXER:**

Multiplexer means transmitting a large number of information units over a smaller number of channels or lines. A digital multiplexer is a combinational circuit that selects binary information from one of many input lines and directs it to a single output line. The selection of a particular input line is controlled by a set of selection lines. Normally there are $2^n$ input line and n selection lines whose bit combination determine which input is selected.

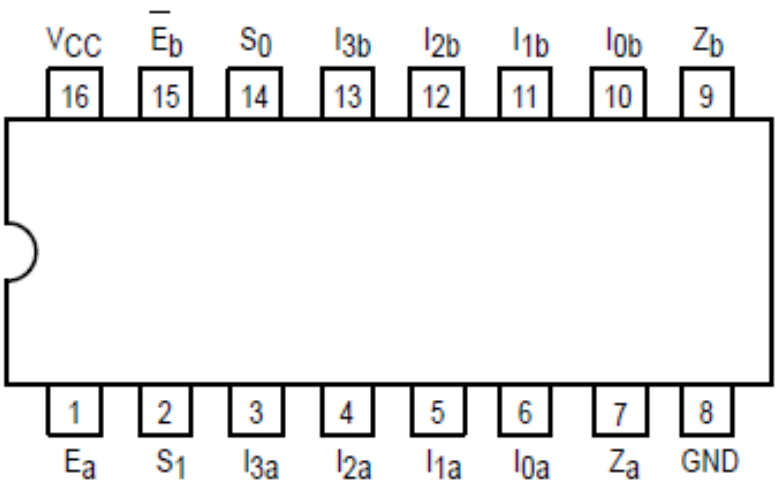
**TRUTH TABLE:**

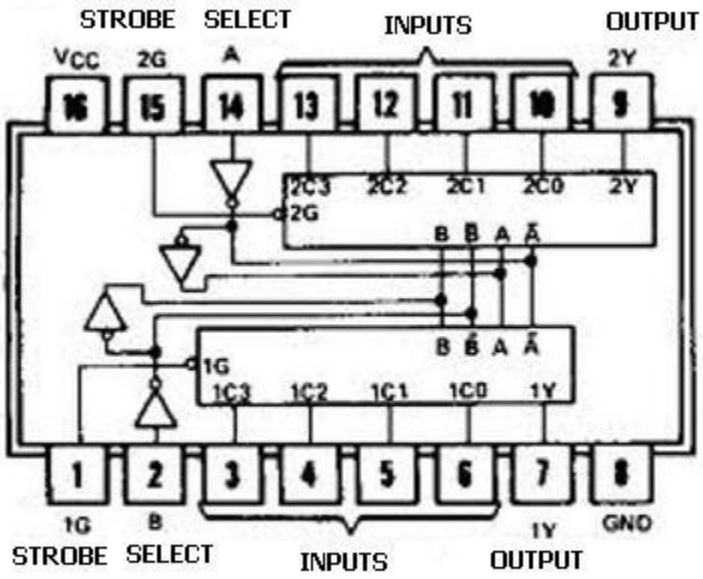
| S1 | S0 | Y = OUTPUT |
|---|---|---|
| 0 | 0 | D0 |
| 0 | 1 | D1 |
| 1 | 0 | D2 |
| 1 | 1 | D3 |

**Equation in SOP form from above table is**

**Y = D0 S1' S0' + D1 S1' S0 + D2 S1 S0' + D3 S1 S0**

**PIN DIAGRAM FOR IC 74153:**




74153

yy

**FUNCTIONAL DESCRIPTION: The'153 is a dual 4-input multiplexer. It can select two bits of data from up to four sources under the control of the common Select inputs (So, Si). The two 4-input multiplexer circuits have individual active LOW Enables (Eabar, Ebbar) which can be used to strobe the outputs independently. When the Enables (E'a, E'b) are HIGH, the corresponding outputs (Za, Zb) are forced LOW. The '153 is the logic implementation of a 2-pole, 4-position switch, where the position of the switch is determined by the logic levels supplied to the two Select inputs. The logic equations for the outputs are shown below.**

**Za = Ea • (loa • Si • So + Ha • Si • So + l2a • Si • So + l3a • Si • So)**
**Zb = Eb • (lob • Si • So + Mb • Si • So + l2b • Si • So + l3b • Si • So)**

**The ’153 can be used to move data from a group of registers to a common output bus. The particular register from which the data came would be determined by the state of the Select inputs. A less obvious application is a function generator. The ’153 can generate two functions of three variables. This is useful for implementing highly irregular random logic.**

**CONNECTION EXAMPLE:**

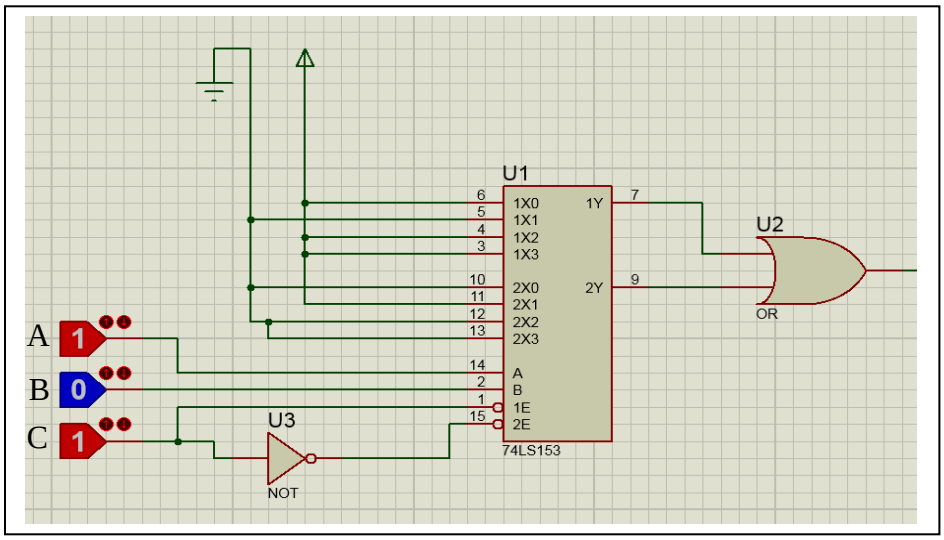
Design following functions using 74153

1) Implement $f(A,B) = \sum(0,2,3,5)$ using single 8:1
2) Implement $f(A,B,C) = \sum(1,2,5,6)$ using single 4:1
3) Implement $f(X,Y,Z) = \sum(0,1,3,7)$ using 8:1
4) Implement $f(I,J,K,L) = \sum(0,2,3,7,11,14,15)$ using 8:1

## Implement $\mathbf{f(C,B,A) = \sum(0,2,3,5)}$

### Truth Table and Circuit Diagram:

| C | B | A | Y | I |
|---|---|---|---|---|
| 0 | 0 | 0 | 1 | I0a |
| 0 | 0 | 1 | 0 | I1a |
| 0 | 1 | 0 | 1 | I2a |
| 0 | 1 | 1 | 1 | I3a |
| 1 | 0 | 0 | 0 | I0b |
| 1 | 0 | 1 | 1 | I1b |
| 1 | 1 | 0 | 0 | I2b |
| 1 | 1 | 1 | 0 | I3b |




**PROCEDURE:**

(i) Connections are given as per circuit diagram.

(ii) Logical inputs are given as per circuit diagram.

(iii) Observe the output and verify the truth table.

**CONCLUSION:**

## EXPERIMENT NO. 4

## STUDY OF DEMULTIPLEXER IC 74138

**AIM :** Study of IC-74LS138 as a Demultiplexer/ Decoder

1. Design and implement full adder and subtractor function using IC-74LS138.
2. Design & Implement 3-bit code converter using IC-74LS138.(Gray to Binary/Binary to Gray)

**APPARATUS REQUIRED:**

| SR.No. | COMPONENT | SPECIFICATION | QTY. |
|---|---|---|---|
| 1. | 1:8 DEMUX | IC 74138 | 1 |
| 2. | NAND GATE | IC 7408 | 1 |
| 3. | OR GATE | IC 7432 | 1 |
| 4. | IC TRAINER KIT | - | 1 |
| 5. | PATCH CORDS | - | 30 |

**THEORY:**

A conversion circuit must be inserted between the two systems if each uses different codes for same information. Thus, code converter is a circuit that makes the two systems compatible even though each uses different binary code.

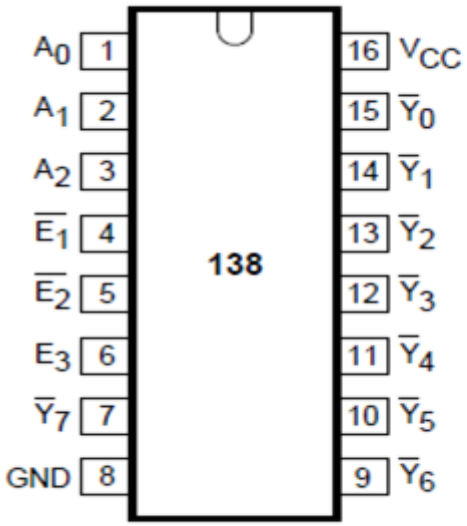
The bit combination assigned to binary code to gray code. Since each code uses three bits to represent a decimal digit. There are three inputs and three outputs. Gray code is a non-weighted code.

The input variable are designated as B2, B1, B0 and the output variables are designated as C2, C1, Co. from the truth table, combinational circuit is designed.

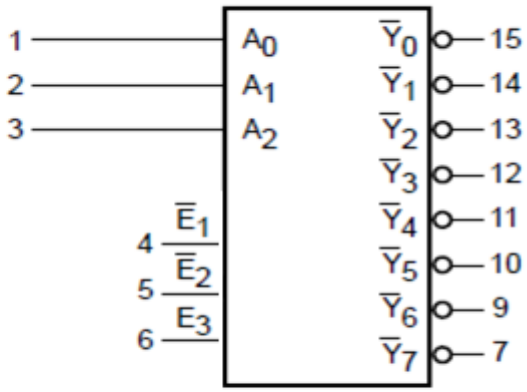
From truth tables functions for each output bit is obtained and implemented using demultiplexer or decoder,

**LOGIC DIAGRAM:**

**PIN DIAGRAM**




Pin configuration.




Logic symbol.

1. **BINARY TO GRAY:**

**TRUTH TABLE:**

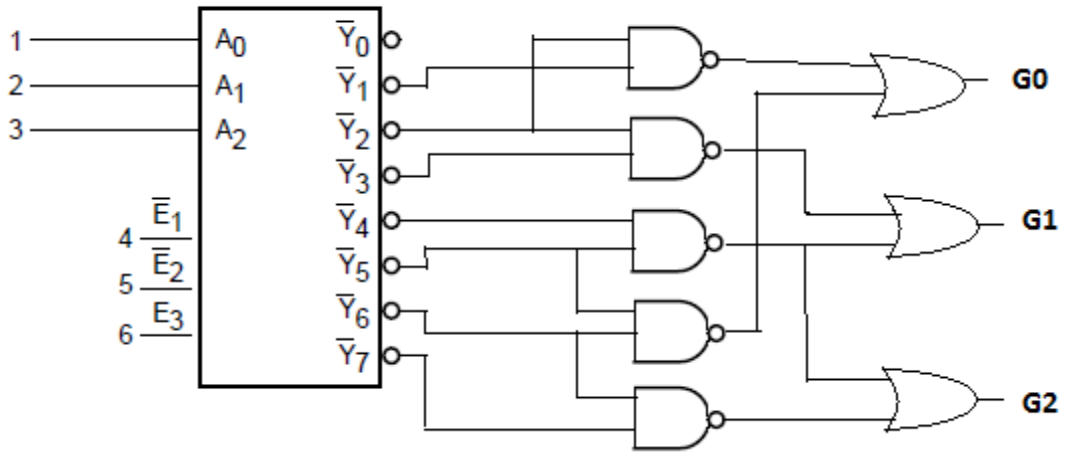
| **Binary input** | | | **Gray code output** | | |
|---|---|---|---|---|---|
| **B2** | **B1** | **B0** | **G2** | **G1** | **G0** |
| **0** | **0** | **0** | **0** | **0** | **0** |
| **0** | **0** | **1** | **0** | **0** | **1** |
| **0** | **1** | **0** | **0** | **1** | **1** |
| **0** | **1** | **1** | **0** | **1** | **0** |
| **1** | **0** | **0** | **1** | **1** | **0** |
| **1** | **0** | **1** | **1** | **1** | **1** |
| **1** | **1** | **0** | **1** | **0** | **1** |
| **1** | **1** | **1** | **1** | **0** | **0** |

$$G2 = \sum m\,(4,5,6,7)$$

$$G1 = \sum m\,(2,3,4,5)$$

$$G0 = \sum m\,(1,2,5,6)$$

**CONNECTION DIAGRAM:**




**2. GRAY TO BINARY:**

**TRUTH TABLE:**

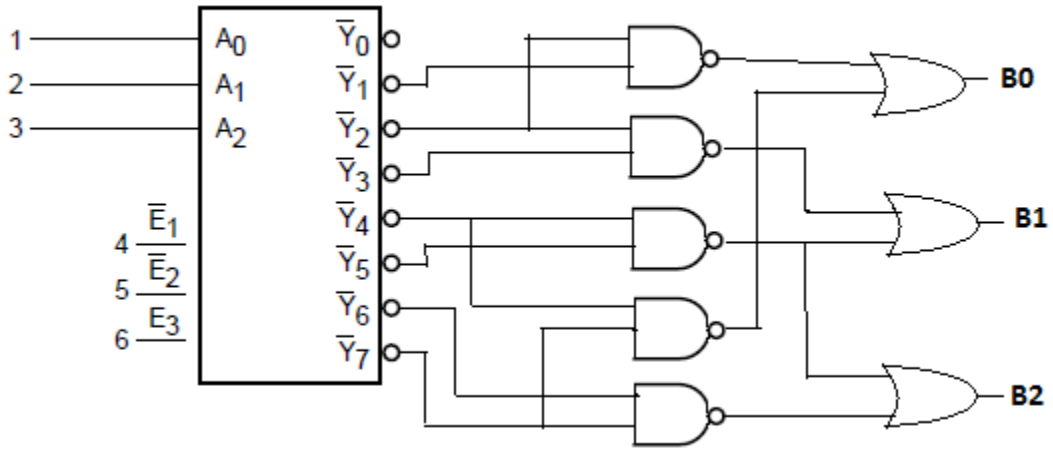
| **Gray Code** | | | **Binary Code** | | |
|---|---|---|---|---|---|
| **G2** | **G1** | **G0** | **B2** | **B1** | **B0** |
| **0** | **0** | **0** | **0** | **0** | **0** |
| **0** | **0** | **1** | **0** | **0** | **1** |
| **0** | **1** | **0** | **0** | **1** | **1** |
| **0** | **1** | **1** | **0** | **1** | **0** |
| **1** | **0** | **0** | **1** | **1** | **1** |
| **1** | **0** | **1** | **1** | **1** | **0** |
| **1** | **1** | **0** | **1** | **0** | **0** |
| **1** | **1** | **1** | **1** | **0** | 1 |

$$B2 = \sum m\,(4,5,6,7)$$

$$B1 = \sum m\,(2,3,4,5)$$

$$B0 = \sum m\,(1,2,4,7)$$

**CONNECTION DIAGRAM:**




**PROCEDURE:**

(i) Connections were given as per circuit diagram.

(ii) Logical inputs were given as per truth table

(iii) Observe the logical output and verify with the truth tables.

**CONCLUSION:**

## EXPERIMENT NO. 5

## DESIGN OF ADDER AND SUBTRACTOR AND BCD ADDER USING 7483

**AIM :** Study of IC-74LS83 as a BCD adder.

1. Design and Implement half adder, subtractor, full adder, subtractor using gates.
2. Design and Implement 1 digit BCD adder using IC-74LS83.
3. Design and Implement 4-bit Binary adder and subtractor using IC-74LS83.

**APPARATUS REQUIRED:**

| Sl.No. | COMPONENT | SPECIFICATION | QTY. |
|---|---|---|---|
| 1. | AND GATE | IC 7408 | 1 |
| 2. | X-OR GATE | IC 7486 | 1 |
| 3. | NOT GATE | IC 7404 | 1 |
| 4. | OR GATE | IC 7432 | 1 |
| 5. | BCD ADDER | IC 7483 | 2 |
| 3. | IC TRAINER KIT | - | 1 |
| 4. | PATCH CORDS | - | 40 |

**THEORY:**

**HALF ADDER:**

A half adder has two inputs for the two bits to be added and two outputs one from the sum ‘ S’ and other from the carry ‘ c’ into the higher adder position. Above circuit is called as a carry signal from the addition of the less significant bits sum from the X-OR Gate the carry out from the AND gate.

**FULL ADDER:**

A full adder is a combinational circuit that forms the arithmetic sum of input; it consists of three inputs and two outputs. A full adder is useful to add three bits at a time but a half adder cannot do so. In full adder sum output will be taken from X-OR Gate, carry output will be taken from OR Gate.

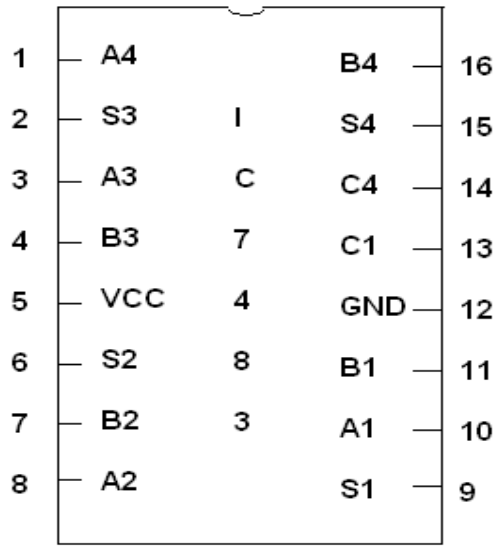
**HALF SUBTRACTOR:**

The half subtractor is constructed using X-OR and AND Gate. The half subtractor has two input and two outputs. The outputs are difference and borrow. The difference can be applied using X-OR Gate, borrow output can be implemented using an AND Gate and an inverter.

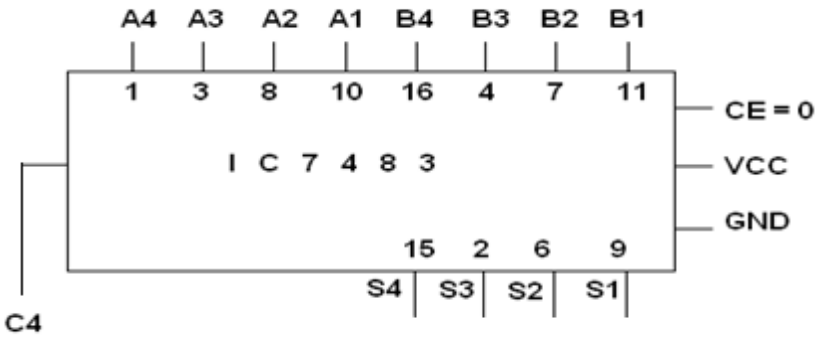
**FULL SUBTRACTOR:**

The full subtractor is a combination of X-OR, AND, OR, NOT Gates. In a full subtractor the logic circuit should have three inputs and two outputs. The two half subtractor put together gives a full subtractor .The first half subtractor will be C and A B. The output will be difference output of full subtractor. The expression AB assembles the borrow output of the half subtractor and the second term is the inverted difference output of first X-OR.

**PIN DIAGRAM FOR IC 7483:**

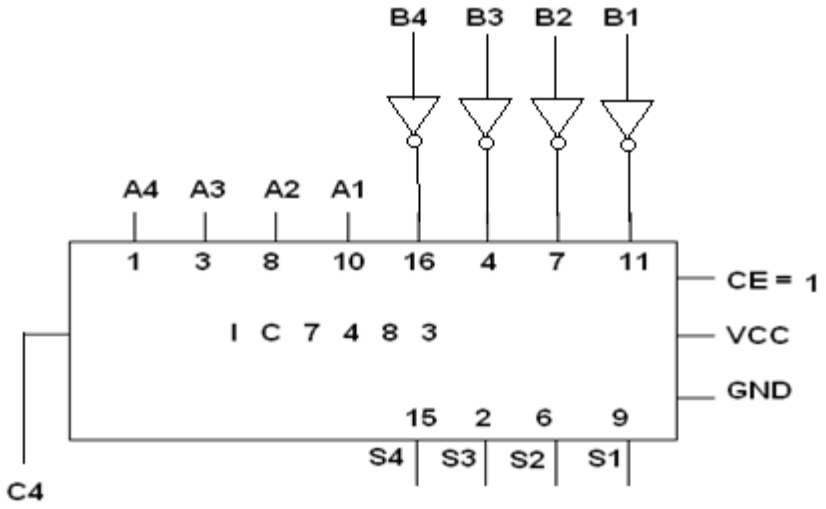



**4 BIT ADDER:**




**4 BIT SUBTRACTOR:**

In 4 bit subtractor 2's complement method is used so CE is connected to VCC or Logic 1 and A-B is possible by taking 1's complement of B using NOT gate.




**4 BIT BCD ADDER:**

Consider the arithmetic addition of two decimal digits in BCD, together with an input carry from a previous stage. Since each input digit does not exceed 9, the output sum cannot be greater than 19, the 1 in the sum being an input carry. The output of two decimal digits must be represented in BCD and should appear in the form listed in the columns.

A BCD adder, which adds 2 BCD digits and produce a sum digit in BCD. The 2 decimal digits, together with the input carry, are first added in the top 4 bit adder to produce the binary sum.

**TRUTH TABLE:**

| BINARY SUM | BCD SUM | | | | | Ten's place |
|---|---|---|---|---|---|---|
| | C4 | S4 | S3 | S2 | S1 | Y |
| 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 0 | 1 | 0 |
| 2 | 0 | 0 | 0 | 1 | 0 | 0 |
| 3 | 0 | 0 | 0 | 1 | 1 | 0 |
| 4 | 0 | 0 | 1 | 0 | 0 | 0 |
| 5 | 0 | 0 | 1 | 0 | 1 | 0 |
| 6 | 0 | 0 | 1 | 1 | 0 | 0 |
| 7 | 0 | 0 | 1 | 1 | 1 | 0 |
| 8 | 0 | 1 | 0 | 0 | 0 | 0 |
| 9 | 0 | 1 | 0 | 0 | 1 | 0 |
| 10 | 0 | 1 | 0 | 1 | 0 | 1 |
| 11 | 0 | 1 | 0 | 1 | 1 | 1 |

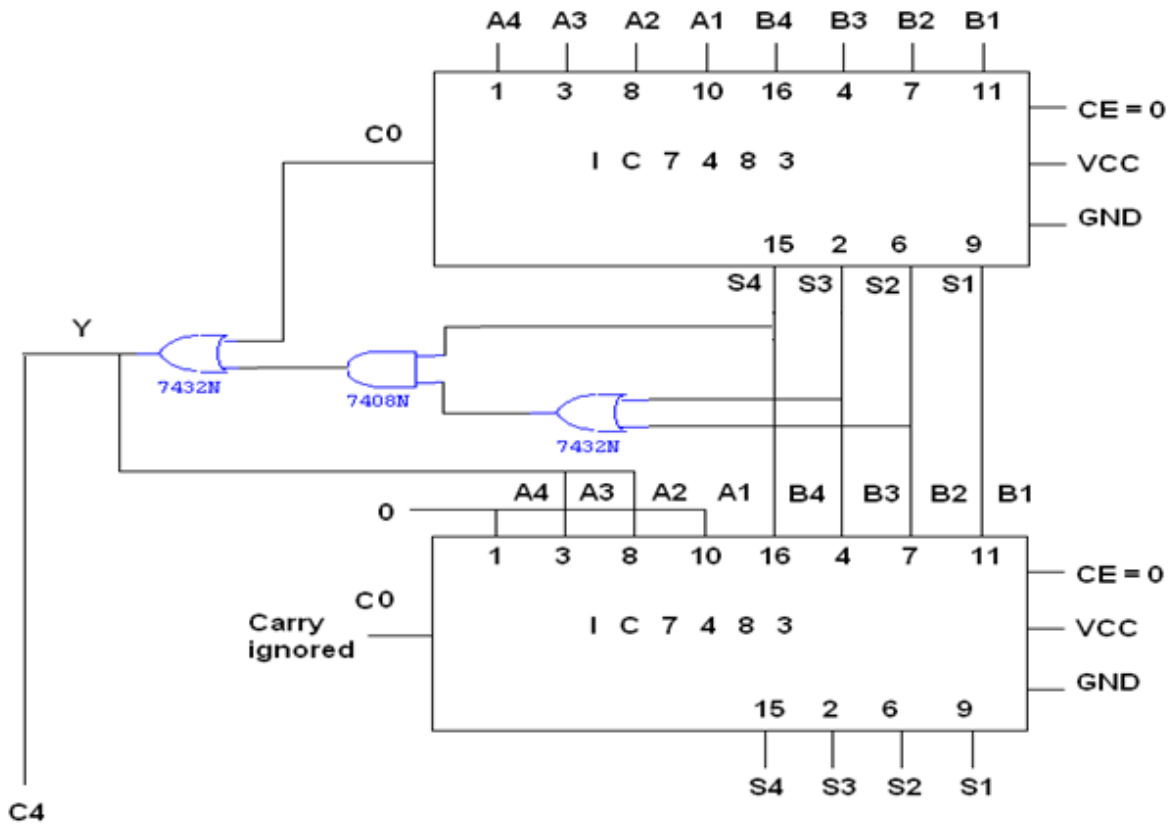
| | | | | | | |
|---|---|---|---|---|---|---|
| **12** | **0** | **1** | **1** | **0** | **0** | **1** |
| **13** | **0** | **1** | **1** | **0** | **1** | **1** |
| **14** | **0** | **1** | **1** | **1** | **0** | **1** |
| **15** | **0** | **1** | **1** | **1** | **1** | **1** |
| **16** | **1** | **0** | **0** | **0** | **0** | **1** |
| **17** | **1** | **0** | **0** | **0** | **1** | **1** |
| **18** | **1** | **0** | **0** | **1** | **0** | **1** |

For sum less than 9 ten's place is zero, because it is single digit and it requires 4 bits for representation.
Sum is greater than 9, i.e. from 10 to 18 ten's place is one and it requires 8 bits for representation.
For ten's place to be one, when sum is 16 to 18 C4 is one and when sum is 10 to 15 S4 is one and S3 or S2 is one.
i.e. C4 OR S4 AND S3 OR S2,

**Y = C4+S4 (S3 + S2)**

LOGIC DIAGRAM:

**BCD ADDER**



**PROCEEDURE:**

(i) Connections are given as per circuit diagram.

(ii) Logical inputs are given as per circuit diagram.

(iii) Observe the output and verify the truth table.

**CONCLUSION:**

**EXPERIMENT NO. 6**

## COMPARATOR

**AIM :** Study of IC-74LS85 as a magnitude comparator.

1. Design and Implement 4-bit Comparator.
2. Design and Implement 8-bit Comparator

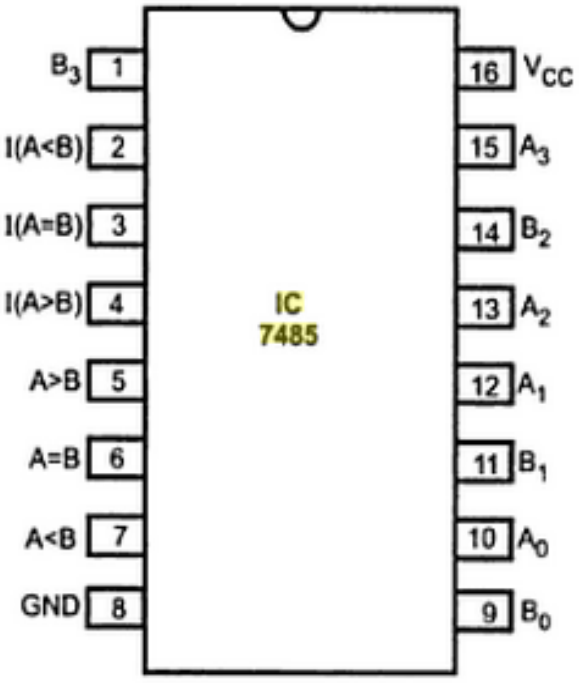
**APPARATUS REQUIRED:**

| Sl.No | COMPONENT | SPECIFICATION | QTY. |
|---|---|---|---|
| 1. | 4 BIT COMPARATOR | IC 7485 | 1 |
| 2. | IC TRAINER KIT | - | 1 |
| 3. | PATCH CORDS | - | 30 |

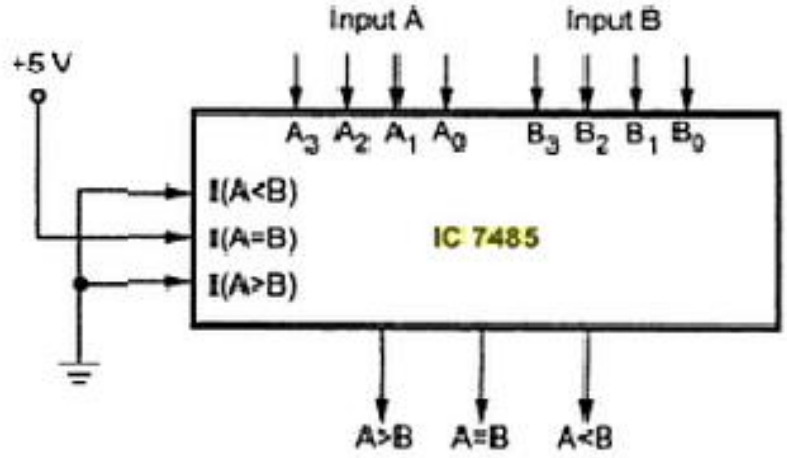
**THEORY:**

**ABOUT COMPARATOR**

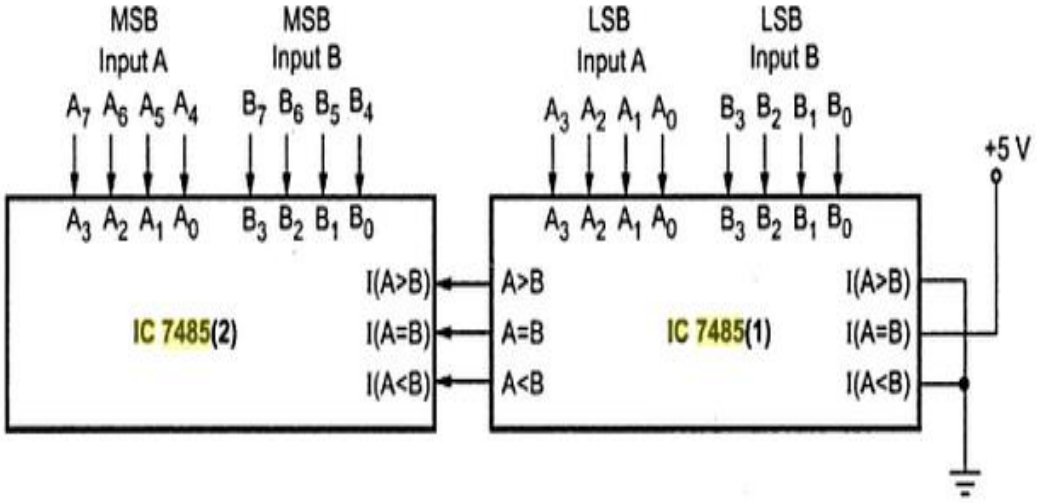
**PIN DIAGRAM FOR IC 7485:**



**4 BIT COMPARATOR:**




**8 BIT COMPARATOR:**




**PROCEEDURE:**

(i) Connections are given as per circuit diagram.

(ii) Logical inputs are given as per circuit diagram.

(iii) Observe the output and verify the truth table.

**CONCLUSION:**

**EXPERIMENT NO. 7**

## ASYNCHRONOUS IC COUNTER

**AIM** **:** Study of Counter ICs (74LS90/74LS93).

1. Design and Implement MOD-6 and MOD-42 using IC-74LS90 and draw Timing Diagram.
2. Design and Implement MOD-7 and MOD-19 using IC-74LS93 and draw Timing Diagram

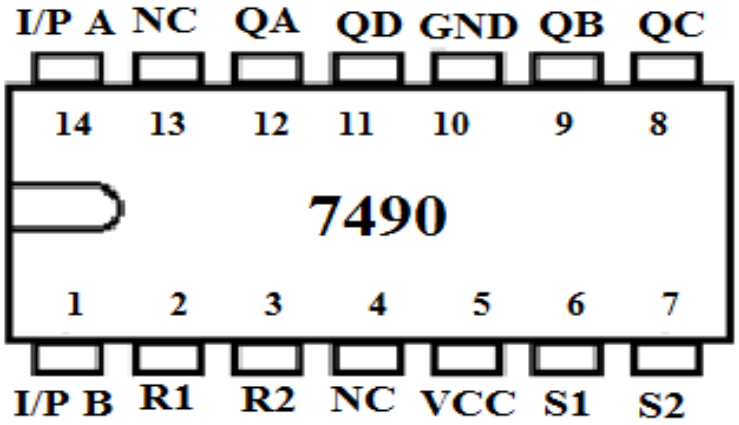
**APPARATUS REQUIRED:**

| Sl.No | COMPONENT | SPECIFICATION | QTY. |
|---|---|---|---|
| 1. | BCD COUNTER | IC 7490 | 2 |
| 2. | BINARY COUNTER | IC 7493 | 2 |
| 3. | AND GATE | IC 7408 | 1 |
| 4. | IC TRAINER KIT | - | 1 |
| 5. | PATCH CORDS | - | 30 |

**THEORY:**

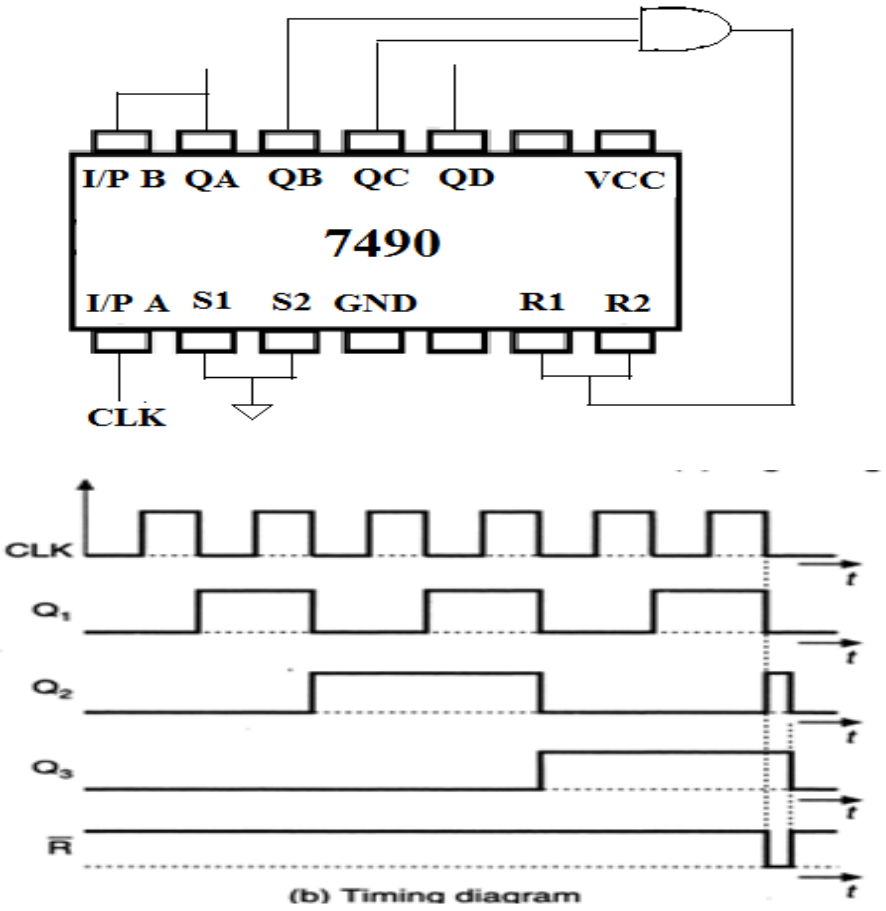
**ABOUT 7490 AND 7493**

**PIN DIAGRAM FOR IC 7490:**



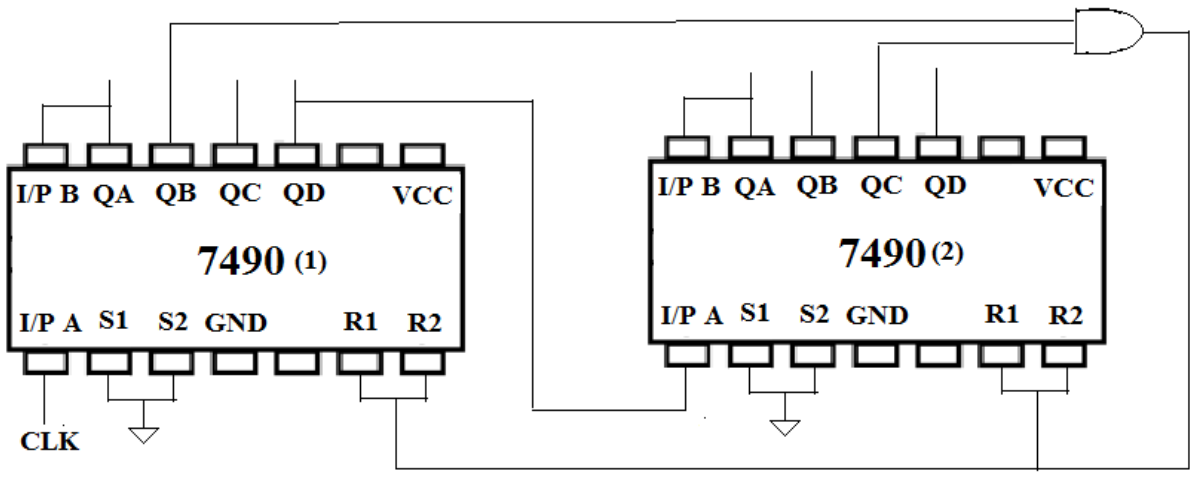

**MOD 6:**

Mod 6 means 0 to 5 states and for 0110 at the output counter should reset.



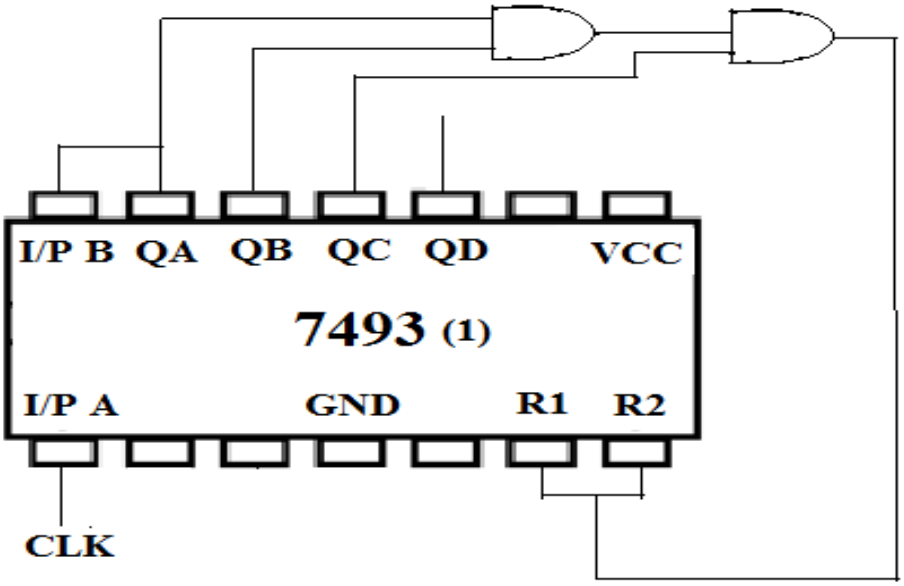

**MOD 42:**

Mod 42 means 0 to 41 states and for 0100 0010 at the output counter should reset.



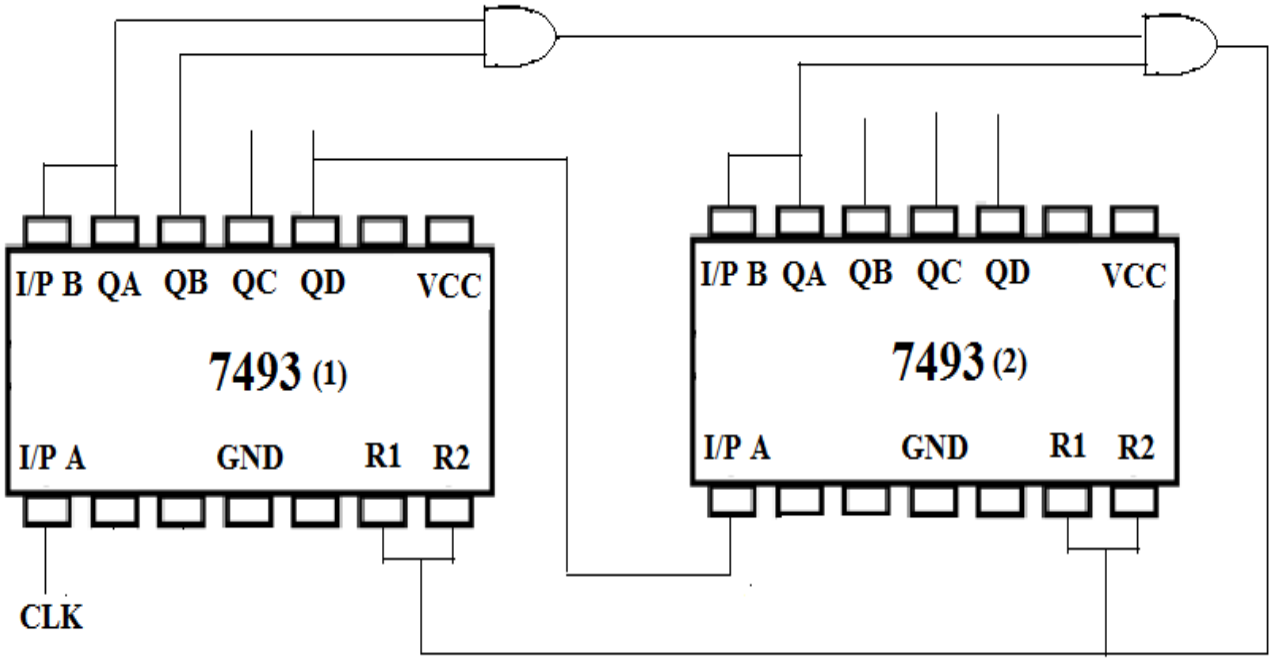

**MOD 7:**

Mod 7 means 0 to 6 states and for 0111 at the output counter should reset.




**MOD 19:**

Mod 19 means 0 to18 states and for binary 19 that is 10011 at the output counter should reset.



**PROCEEDURE:**

(i) Connections are given as per circuit diagram.

(ii) Logical inputs are given as per circuit diagram.

(iii) Observe the output and verify the truth table.

**CONCLUSION:**

## EXPERIMENT NO. 8

## SYNCHRONOUS IC COUNTER

**AIM :** Study of Counter ICs (74LS191).

1. Design & Implement 4-bit Up/down Counter.
2. MOD-11 and MOD-9 Up/down Counter using IC-74HC191. Draw Timing Diagram

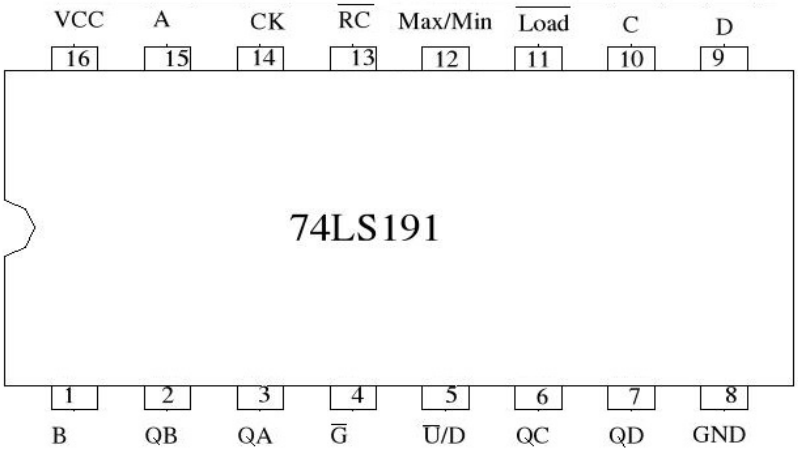
**APPARATUS REQUIRED:**

| Sl.No | COMPONENT | SPECIFICATION | QTY. |
|---|---|---|---|
| 1. | BINARY COUNTER | IC 74191 | 1 |
| 2. | IC TRAINER KIT | - | 1 |
| 3. | PATCH CORDS | - | 30 |

**THEORY:**

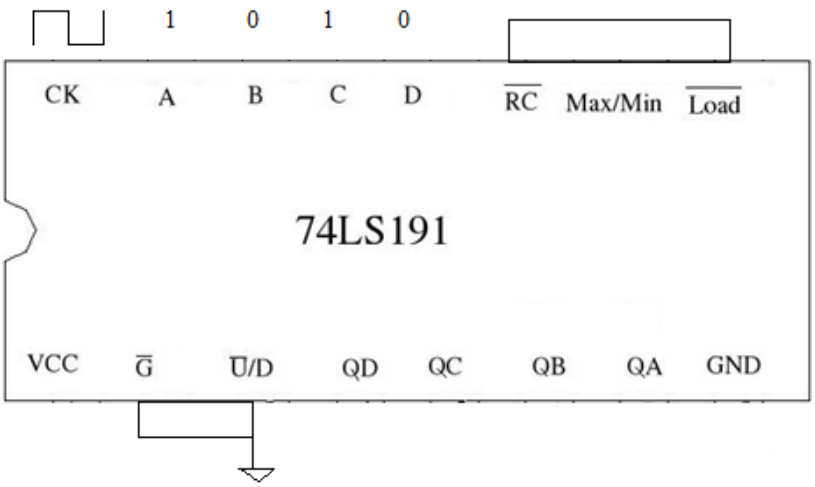
The outputs of the four master-slave flip-flops are triggered on a LOW-to-HIGH level transition of the clock input, if the enable input is LOW. The direction of the count is determined by the level of the down/up input. When LOW, the counter counts up and when HIGH, it counts down. Two outputs have been made available to perform the cascading function that is ripple clock and maximum/minimum count. The ripple clock output produces a low-level output pulse when an overflow or underflow condition exists.

**PIN DIAGRAM FOR IC 74191:**



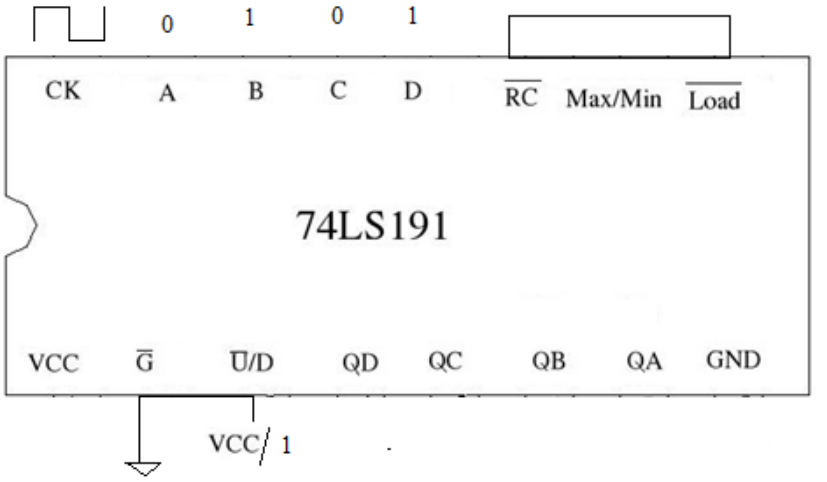

**MOD 11:**

Preset inputs for up counting is 16-11=5 that is 0101. When max count is at the output RC becomes 1 to 0 which is connected to load so counter starts with zero.



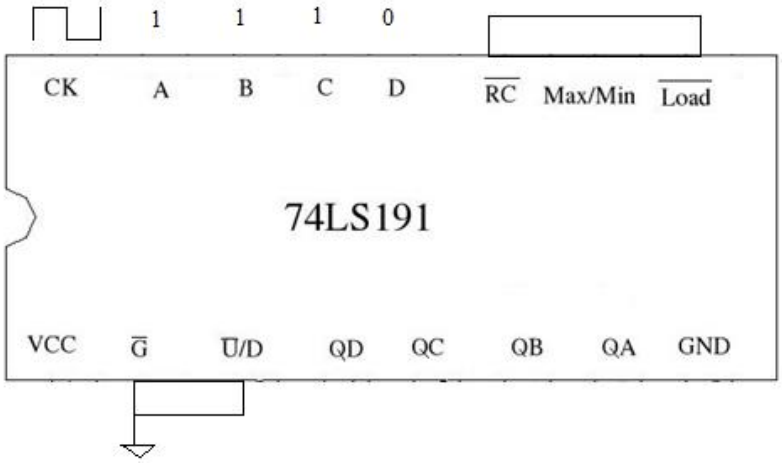

**MOD 11:**

Preset inputs for down counting is 11-1=10 that is 1010. When min count is at the output RC becomes 1 to 0 which is connected to load so counter starts with zero.



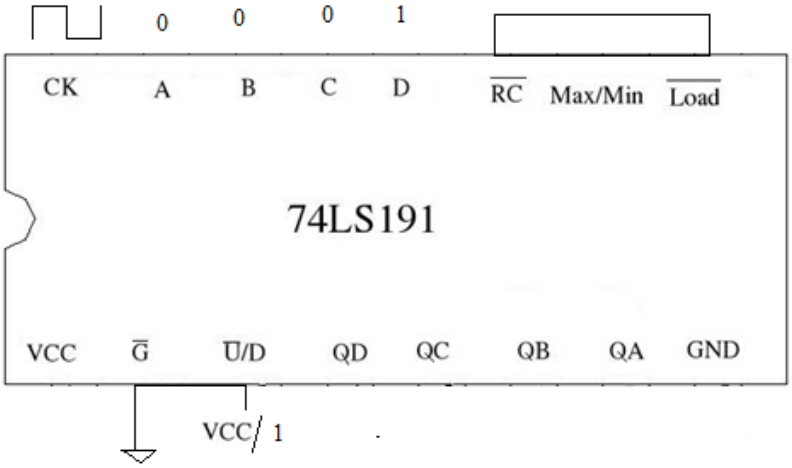

**MOD 9:**

Preset inputs for up counting is 16-9=7 that is 0111.




**MOD 9:**

Preset inputs for down counting is 9-1=8 that is 1000




**PROCEEDURE:**

(i) Connections are given as per circuit diagram.

(ii) Logical inputs are given as per circuit diagram.

(iii) Observe the output and verify the truth table.

**CONCLUSION:**

**EXPERIMENT NO. 9**

**SHIFT REGISTER**

**AIM :** Study of Shift Register (74HC194/74LS95)

1. Design and Implement Pulse train generator using IC-74HC194 (Use right shift/left shift) for sequence 110101.
2. Design and Implement 4-bit Ring Counter/ Twisted ring Counter using shift registers IC 74HC194.

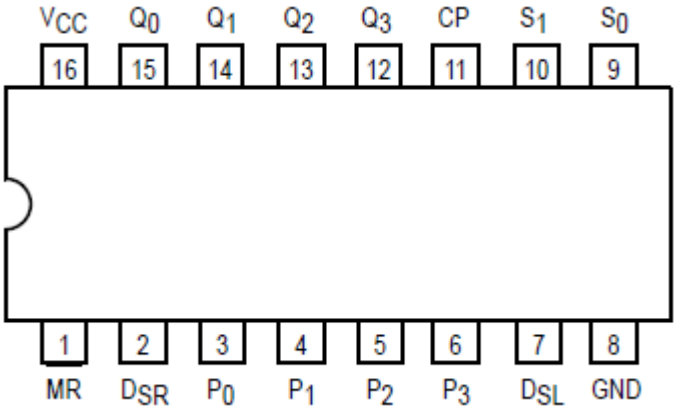
**APPARATUS REQUIRED:**

| Sl.No. | COMPONENT | SPECIFICATION | QTY. |
|---|---|---|---|
| 1. | UNIVERSAL SHIFT REGISTER | IC 74194 | 1 |
| 2. | AND GATE | IC 7408 | 1 |
| 3. | OR GATE | IC 7432 | 1 |
| 4. | NOT GATE | IC 7404 | 1 |
| 5. | IC TRAINER KIT | - | 1 |
| 6. | PATCH CORDS | - | 30 |

**THEORY:**

About ring counter, johnson counter and sequence generator using shift register.

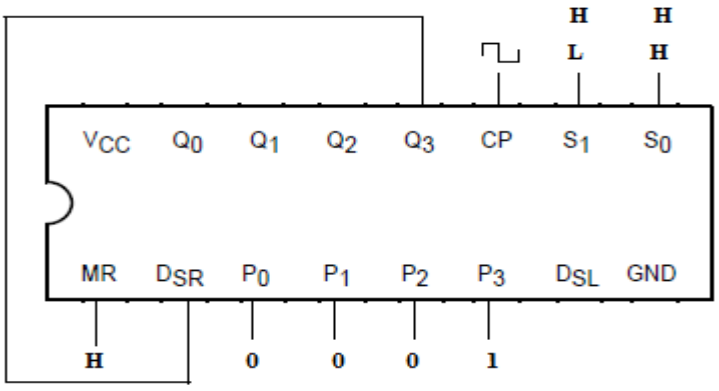
**PIN DIAGRAM:**



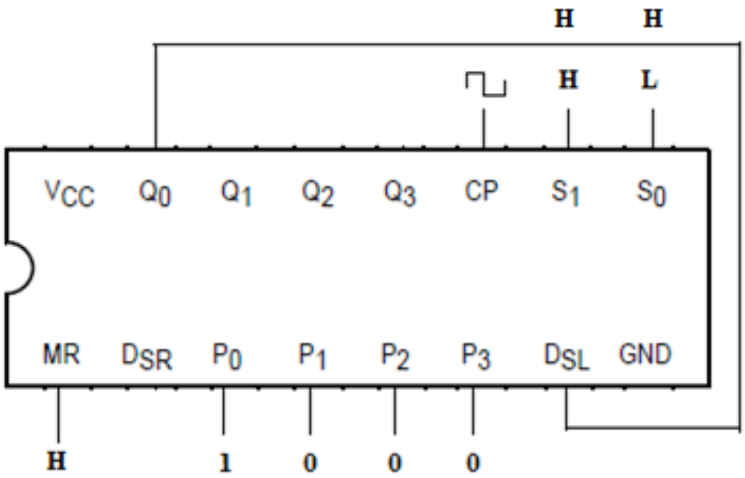
| OPERATING MODE | INPUTS | | | | | | OUTPUTS | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| | MR | $S_1$ | $S_0$ | $D_{SR}$ | $D_{SL}$ | $P_n$ | $Q_0$ | $Q_1$ | $Q_2$ | $Q_3$ |
| Reset | L | X | X | X | X | X | L | L | L | L |
| Hold | H | l | l | X | X | X | $q_0$ | $q_1$ | $q_2$ | $q_3$ |
| Shift Left | H<br>H | h<br>h | l<br>l | X<br>X | l<br>h | X<br>X | $q_1$<br>$q_1$ | $q_2$<br>$q_2$ | $q_3$<br>$q_3$ | L<br>H |
| Shift Right | H<br>H | l<br>l | h<br>h | l<br>h | X<br>X | X<br>X | L<br>H | $q_0$<br>$q_0$ | $q_1$<br>$q_1$ | $q_2$<br>$q_2$ |
| Parallel Load | H | h | h | X | X | $P_n$ | $P_0$ | $P_1$ | $P_2$ | $P_3$ |

**CIRCUIT DIAGRAM:**

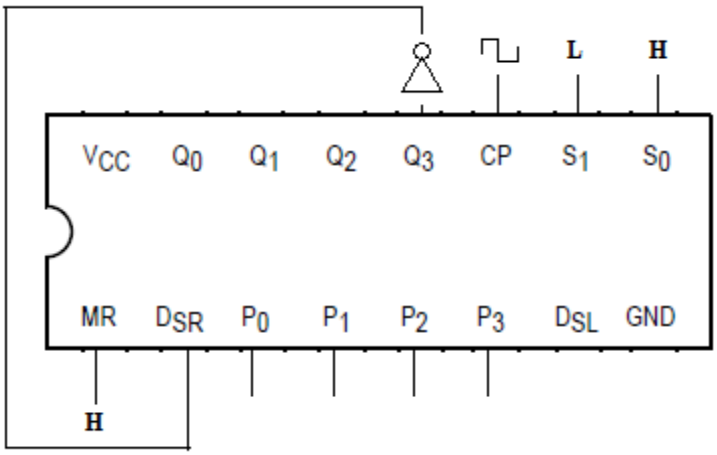
Ring counter right shift



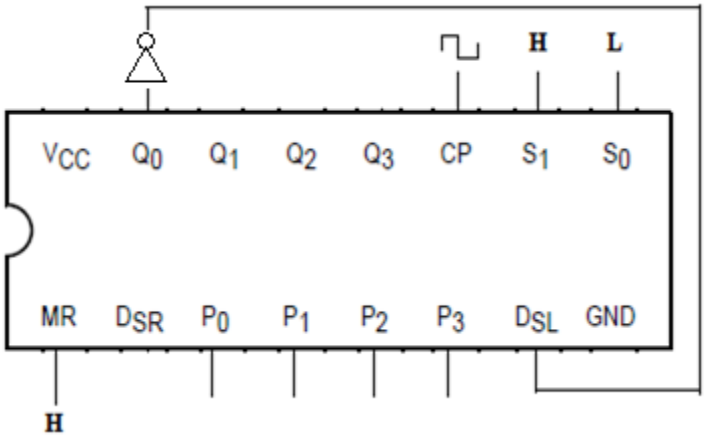

Ring counter left shift



Twisted ring counter right shift:




Twisted ring counter left shift:



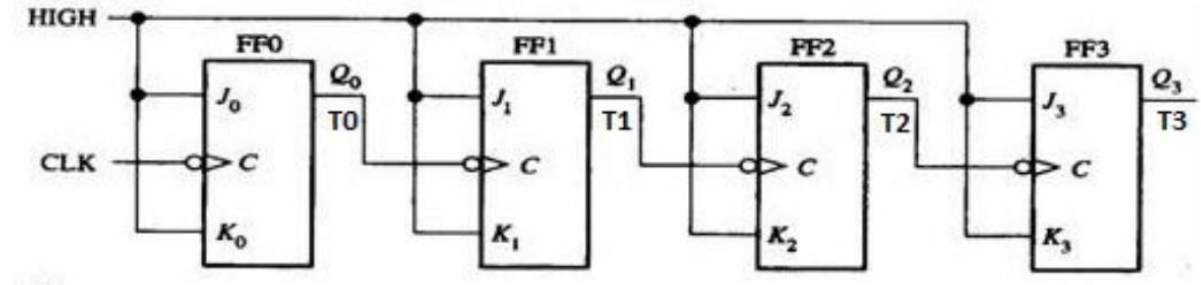

**CONCLUSION:**

**EXPERIMENT NO: 10**

**JK FLIP-FLOP**

**AIM:** Design and Implement 4 bit counter using JK flip Flop.

**THEORY:**



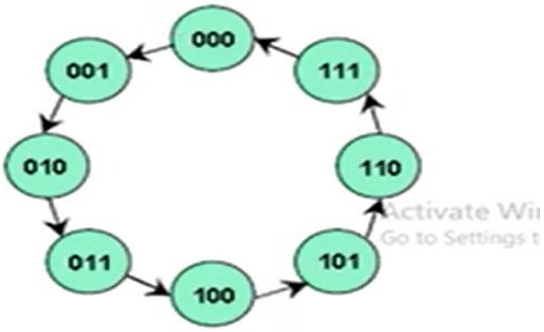

4 bit Binary Counter using JK Flip flop

Steps:




Ex. To design a 3 bit counter

- Let's look at each of the above steps in detail with an example
- Let's say that we are trying to design a 3-bit up counter




1. Decide which flip-flop to use and write down the flip-flop transition table

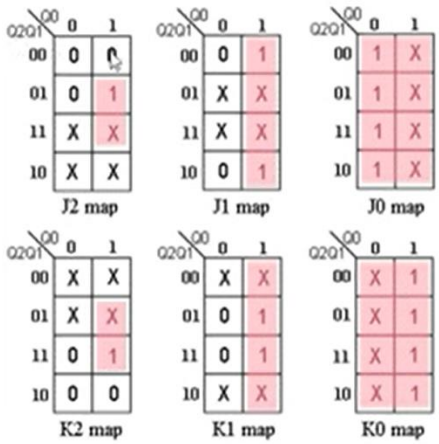
| J | K | Q(t + 1) |
|---|---|---|
| 0 | 0 | Q(t) |
| 0 | 1 | 0 |
| 1 | 0 | 1 |
| 1 | 1 | $\overline{Q}(t)$ |

| Q(t) | Q(t+1) | J | K |
|---|---|---|---|
| 0 | 0 | 0 | × |
| 0 | 1 | 1 | × |
| 1 | 0 | × | 1 |
| 1 | 1 | × | 0 |

## 2. Prepare next state table

| Output State Transitions | | Flip-flop inputs | | |
|---|---|---|---|---|
| Present State $Q_2\ Q_1\ Q_0$ | Next State $Q_2\ Q_1\ Q_0$ | $J_2\ K_2$ | $J_1\ K_1$ | $J_0\ K_0$ |
| 0 0 0 | 0 0 1 | 0 X | 0 X | 1 X |
| 0 0 1 | 0 1 0 | 0 X | 1 X | X 1 |
| 0 1 0 | 0 1 1 | 0 X | X 0 | 1 X |
| 0 1 1 | 1 0 0 | 1 X | X 1 | X 1 |
| 1 0 0 | 1 0 1 | X 0 | 0 X | 1 X |
| 1 0 1 | 1 1 0 | X 0 | 1 X | X 1 |
| 1 1 0 | 1 1 1 | X 0 | X 0 | 1 X |
| 1 1 1 | 0 0 0 | X 1 | X 1 | X 1 |

## 3. Find logical expressions for each flip-flop input using K-map

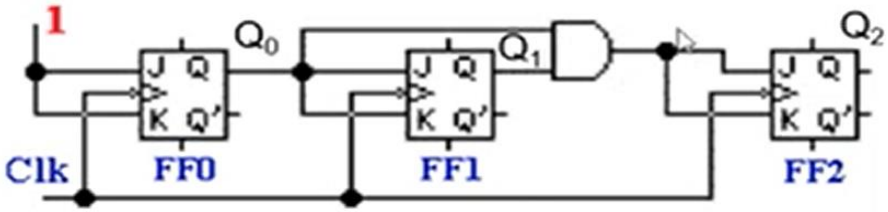



In the K-map, **use the present state of the flip-flop outputs (Q0,Q1,....) as input variables** and **flip-flop input values (e.g. J, K, D, T) as outputs**

$$J_0 = K_0 = 1$$

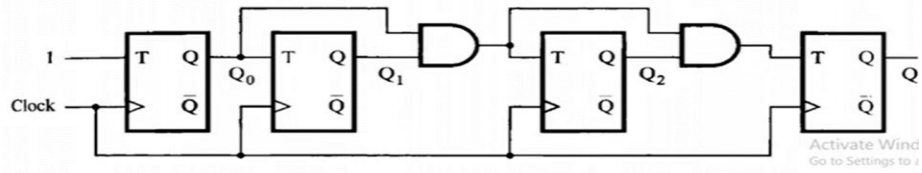
$$J_1 = K_1 = Q_0$$

$$J_2 = K_2 = Q_1 * Q_0$$

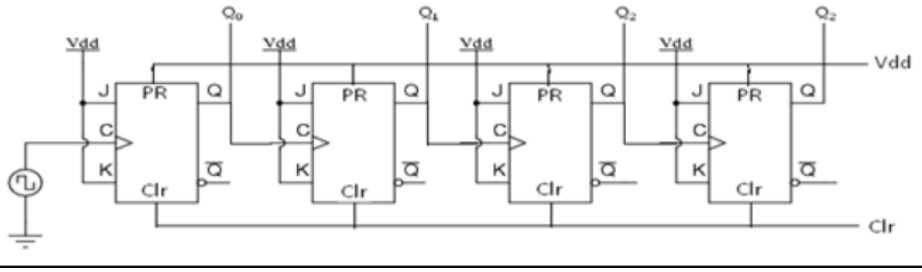



4 bit counter using T Flip Flops:

$$T_0 = 1$$
$$T_1 = Q_0$$
$$T_2 = Q_0Q_1$$
$$T_3 = Q_0Q_1Q_2$$




**Logic Diagram for 4- bit Up Counter:**




## CONCLUSION:

**EXPERIMENT NO: 11**

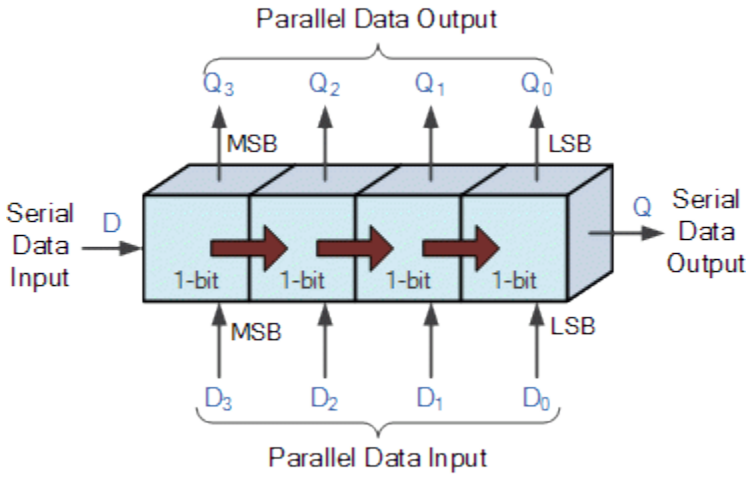
## STUDY OF SHIFT REGISTER

**AIM**: Design and Implement 4-bit right shift and left shift register using D-flip flop.

**THEORY:**

### The Shift Register

The Shift Register is another type of sequential logic circuit that can be used for the storage or the transfer of binary data




This sequential device loads the data present on its inputs and then moves or "shifts" it to its output once every clock cycle, hence the name **Shift Register**.
A *shift register* basically consists of several single bit "D-Type Data Latches", one for each data bit, either a logic "0" or a "1", connected together in a serial type daisy-chain arrangement so that the output from one data latch becomes the input of the next latch and so on.

Data bits may be fed in or out of a shift register serially, that is one after the other from either the left or the right direction, or all together at the same time in a parallel configuration.

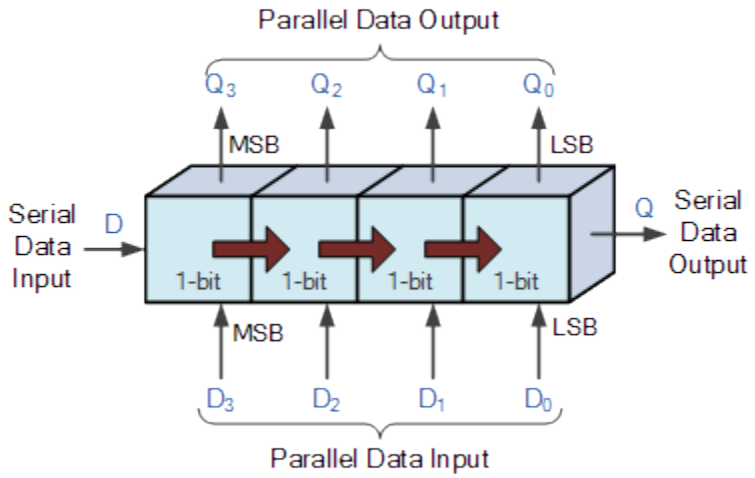
The number of individual data latches required to make up a single **Shift Register** device is usually determined by the number of bits to be stored with the most common being 8-bits (one byte) wide constructed from eight individual data latches.

*Shift Registers* are used for data storage or for the movement of data and are therefore commonly used inside calculators or computers to store data such as two binary numbers before they are added together, or to convert the data from either a serial to parallel or parallel to serial format. The individual data latches that make up a single shift register are all driven by a common clock ( Clk ) signal making them synchronous devices.

Shift register IC's are generally provided with a *clear* or *reset* connection so that they can be "SET" or "RESET" as required. Generally, shift registers operate in one of four different modes with the basic movement of data through a shift register being:

- Serial-in to Parallel-out (SIPO) - the register is loaded with serial data, one bit at a time, with the stored data being available at the output in parallel form.
- Serial-in to Serial-out (SISO) - the data is shifted serially "IN" and "OUT" of the register, one bit at a time in either a left or right direction under clock control.
- Parallel-in to Serial-out (PISO) - the parallel data is loaded into the register simultaneously and is shifted out of the register serially one bit at a time under clock control.
- Parallel-in to Parallel-out (PIPO) - the parallel data is loaded simultaneously into the register, and transferred together to their respective outputs by the same clock pulse.

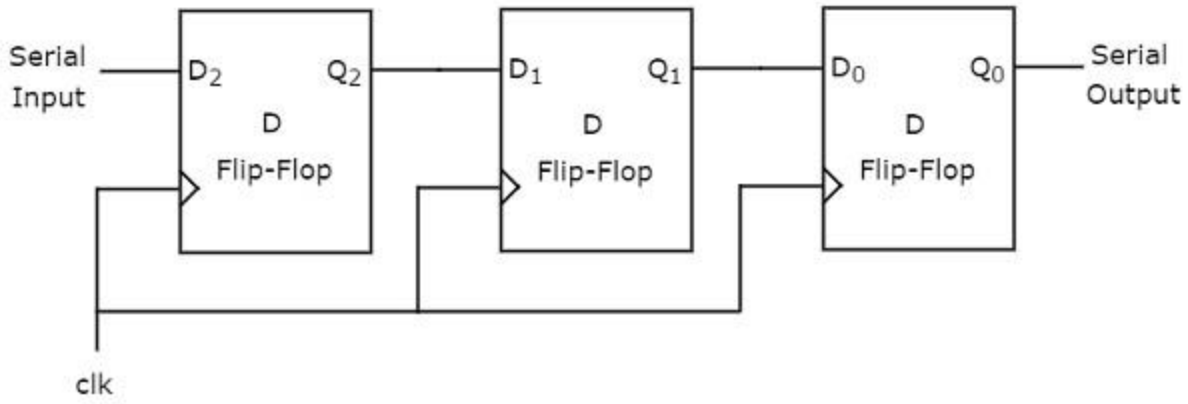
The effect of data movement from left to right through a shift register can be presented graphically as:




Also, the directional movement of the data through a shift register can be either to the left, (left shifting) to the right, (right shifting) left-in but right-out, (rotation) or both left and right shifting within the same register thereby making it *bidirectional*. In this tutorial it is assumed that all the data shifts to the right, (right shifting).

Serial In − Serial Out SISOSISO Shift Register

The shift register, which allows serial input and produces serial output is known as Serial In – Serial Out SISOSISO shift register. The **block diagram** of 3-bit SISO shift register is shown in the following figure.




This block diagram consists of three D flip-flops, which are **cascaded**. That means, output of one D flip-flop is connected as the input of next D flip-flop. All these flip-flops are synchronous with each other since, the same clock signal is applied to each one.

In this shift register, we can send the bits serially from the input of left most D flip-flop. Hence, this input is also called as **serial input**. For every positive edge triggering of clock signal, the data shifts from one stage to the next. So, we can receive the bits serially from the output of right most D flip-flop. Hence, this output is also called as **serial output**.

Example

Let us see the working of 3-bit SISO shift register by sending the binary information **"011"** from LSB to MSB serially at the input.

Assume, initial status of the D flip-flops from leftmost to rightmost is Q2Q1Q0=000Q2Q1Q0=000. We can understand the **working of 3-bit SISO shift register** from the following table.

| No of positive edge of Clock | Serial Input | $Q_2$ | $Q_1$ | $Q_0$ |
|---|---|---|---|---|
| 0 | - | 0 | 0 | 0 |
| 1 | 1LSBLSB | 1 | 0 | 0 |
| 2 | 1 | 1 | 1 | 0 |
| 3 | 0MSBMSB | 0 | 1 | 1LSBLSB |
| 4 | - | - | 0 | 1 |
| 5 | - | - | - | 0MSBMSB |

The initial status of the D flip-flops in the absence of clock signal is $Q_2Q_1Q_0$=000Here, the serial output is coming from $Q_0$. So, the LSB 1 is received at 3rd positive edge of clock and the MSB 0 is received at 5th positive edge of clock.

Therefore, the 3-bit SISO shift register requires five clock pulses in order to produce the valid output. Similarly, the **N-bit SISO shift register** requires **2N-1** clock pulses in order to shift 'N' bit information.

## CONCLUSION: